

CS4800: Designing Sustainable ICT Systems

Lecture 5.2: Sustainable Cloud Computing, Part 2

October 1, 2026

Lecturer: Przemysław Pawełczak

Authors: Alex Nedelcu, Cristian Cuțitei, Przemysław Pawełczak

CS4800-EWI@tudelft.nl

What will you learn in this lecture?

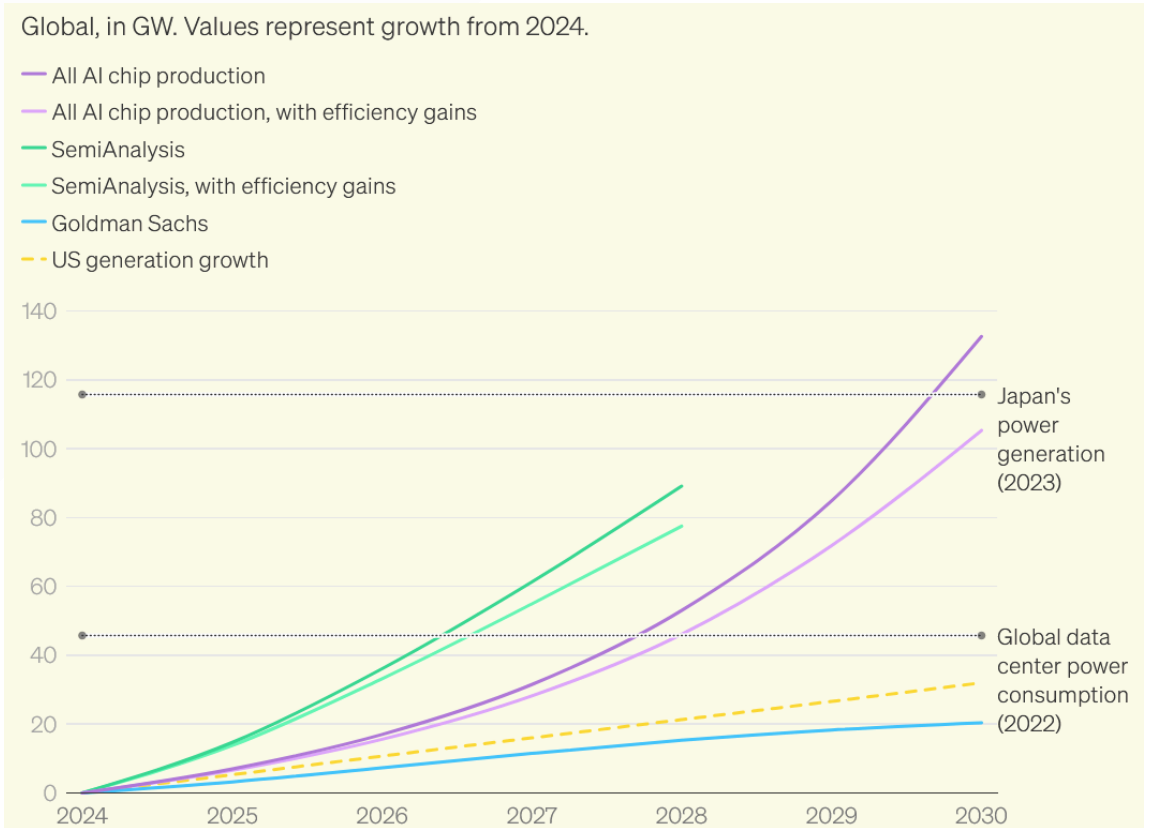
Lecture 5.2: Sustainable Cloud Computing, Part 2

- Industry trends: compute growth, efficiency gains, energy
- Prospective solutions: cloud strategy, demand response, cooling technologies, waste heat reuse

What are the important trends of the data center industry?

Industry trends: power demand

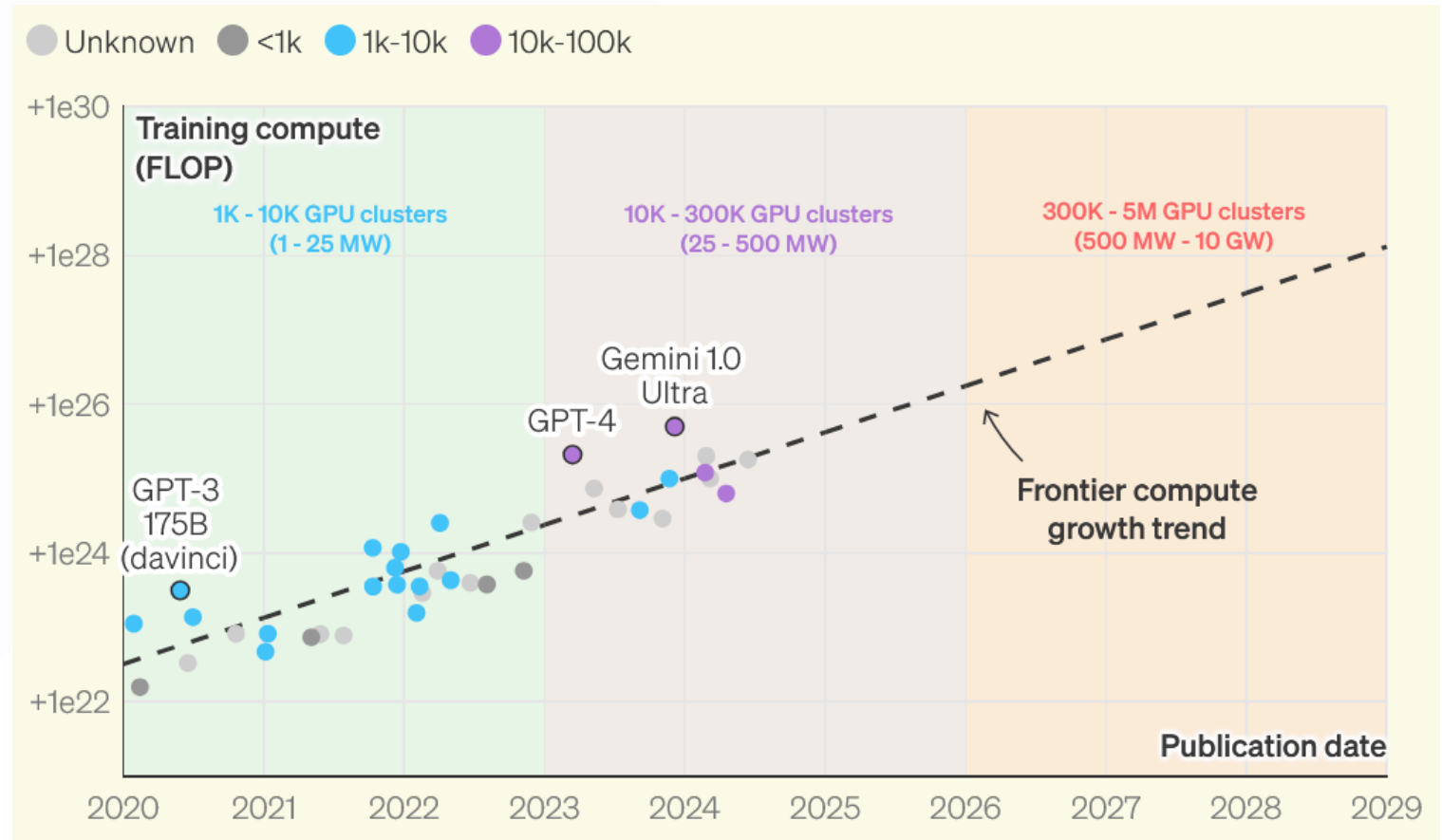
- All estimates: projected increase in global data center power demand: both AI and non-AI ([doubled](#) 2019-2024!)
- [Goldman Sachs](#):
 - non-crypto data center power demand grows 80-240% by 2030
 - 3% of US power demand → 8%
 - 20% of growth is AI
- Other [estimates](#) (e.g. from expected production of AI chips): sharper, exponential growth



[Various estimates for AI-related demand growth](#)

Industry trends: compute demand

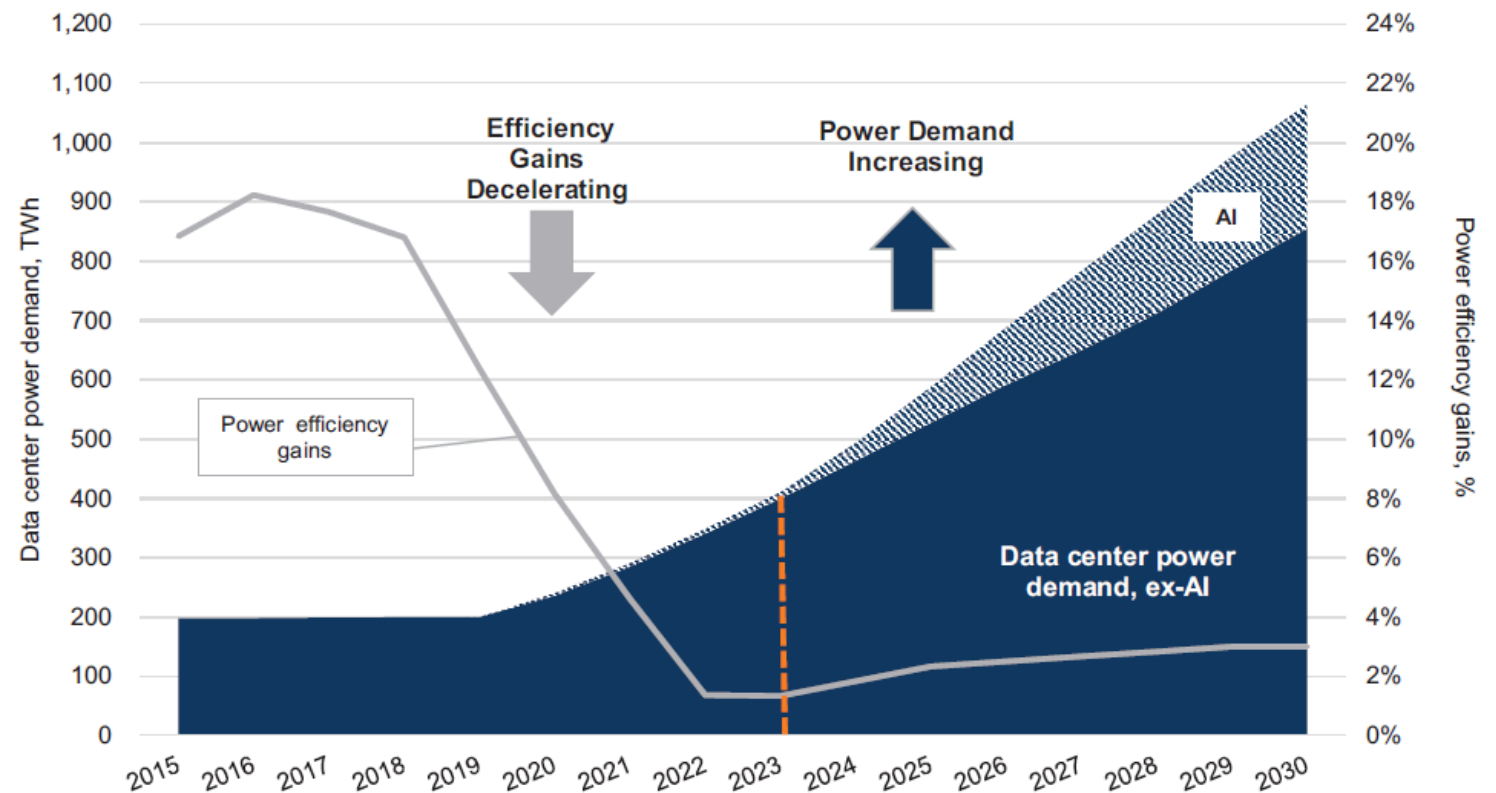
- Divergent power demand estimates: different assumptions on AI growth
- Recent compute growth for frontier AI models: [quintupling scaling law](#) (5x growth per year)
- No consensus on future: hardware improvement, data availability?



[Historic trend for frontier model training compute](#)

Industry trends: efficiency gains

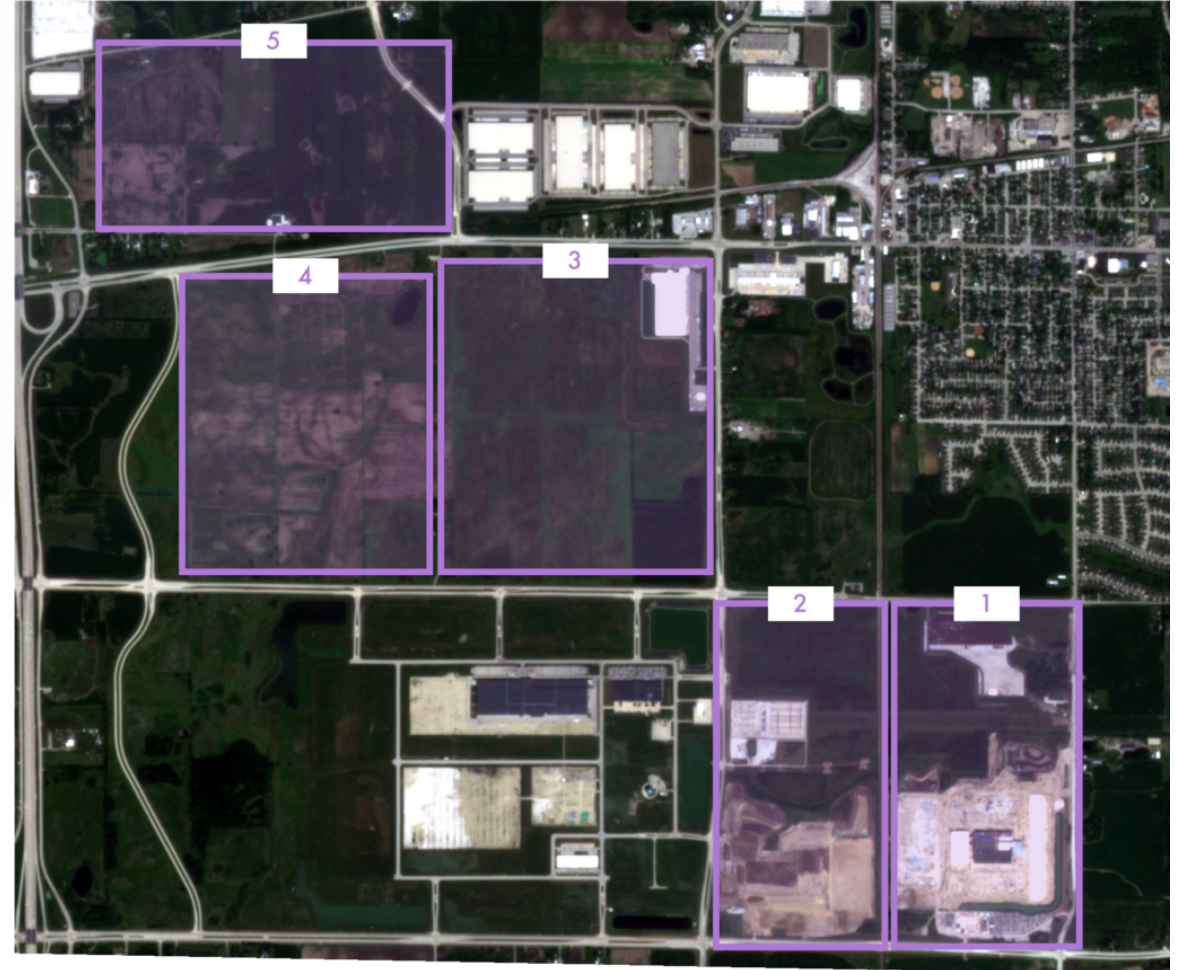
- [2015-2019](#): constant data center power demand, efficiency gains keep up (~15% yearly)
- [Currently](#): efficiency gains down to 2-4% per year
- Even modern systems (e.g. NVIDIA DGX) cannot keep up with AI training/inference demand growth



[Projected data center power demand and efficiency gains](#)

Industry trends: data center size

- Modern data centers: >10,000 GPUs, >10MW
- Under construction: >100MW
- [Plans](#): >1GW
- Planned data centers cannot fit in single building/be powered by single grid → campus with wide-area network (WAN)
 - Distributed data centers unlock more AI training → feedback loop!
 - Global fiber cable manufacturing capacity [partially reserved](#) by builders!



[Planned Microsoft/OpenAI data center campus](#)

Industry trends: energy

- Renewable energy buildout [cannot keep up](#) with data center demand growth
- Coal plants meant to be dismantled [kept online](#) to power new data centers
- [Newly planned](#) buildout of gas plants, pipelines
- Data center builders not waiting for grid interconnection, deploying [trailer-mounted gas](#) turbines 'behind-the-meter'
- This is not good



[xAI data center with 14 mobile gas turbines](#)

What solutions do we have at our disposal?

Prospective solutions: economies of scale

- Hyperscale data centers [more efficient](#) than colocation/internal
- Economies of scale: larger, more energy efficient cooling system vs regular air conditioning; fewer power electronics per computer
- However: hyperscale data center needs to run continuously!
- Also important: where is data center found?
- [Strategic regions](#): desirable geographic/climatic aspects → reduce need for cooling, water



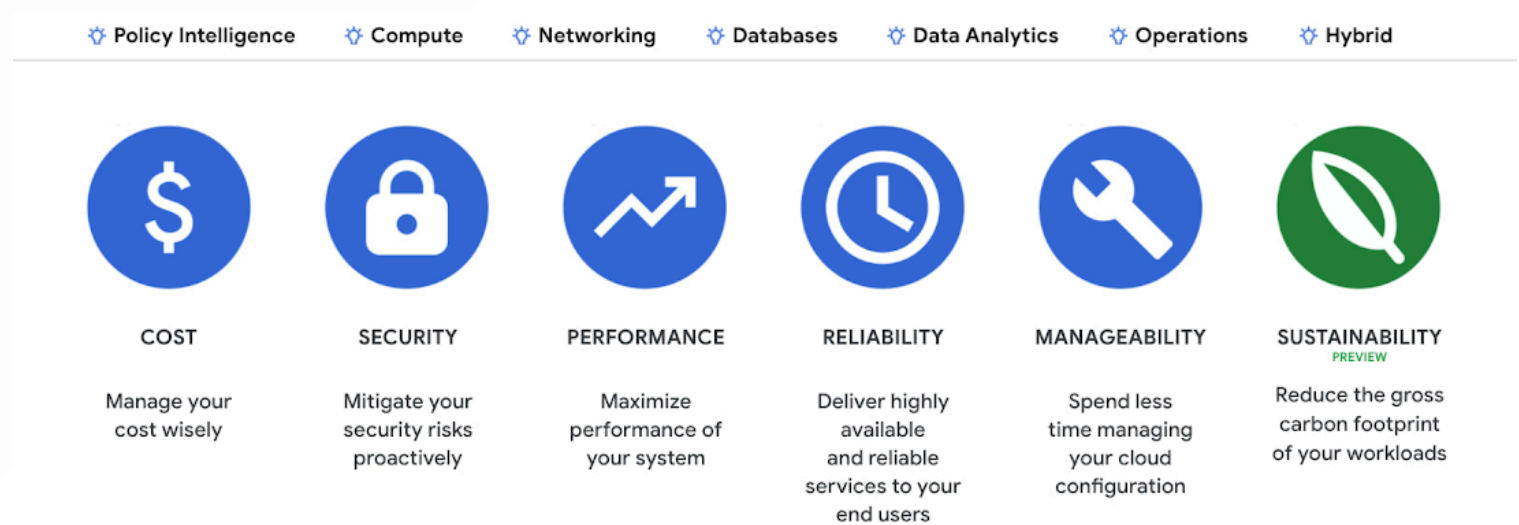
Hyperscale data centres are significantly more efficient than internal data centres

Category	Energy use Million MWh	Computing workloads million	Water intensity M ₃ MWh ₋₁	Carbon intensity ton CO ₂ -eq MWh ₋₁	Water intensity m ₃ /workload	Carbon intensity Ton CO ₂ -eq /workload
Internal	26.90	16	7.20	0.45	12.15	0.75
Colocation	22.4	41	7.00	0.42	3.85	0.25
Hyperscale	22.85	76	7.00	0.44	2.10	0.15

Comparison between hyperscale, colocated, and on-prem data centers

Prospective solutions: cloud strategy

- Power electronics [designed to operate efficiently at maximum load](#) → objective is to [maximize server utilization](#)
- Cloud service providers offer sustainability platforms: emissions tracking, [carbon optimization](#) (Microsoft Azure), smart scheduling (Google Cloud)
- Amazon Web Services (AWS) [shared responsibility model](#):
 - AWS reduces data center impact
 - User reduces workload impact



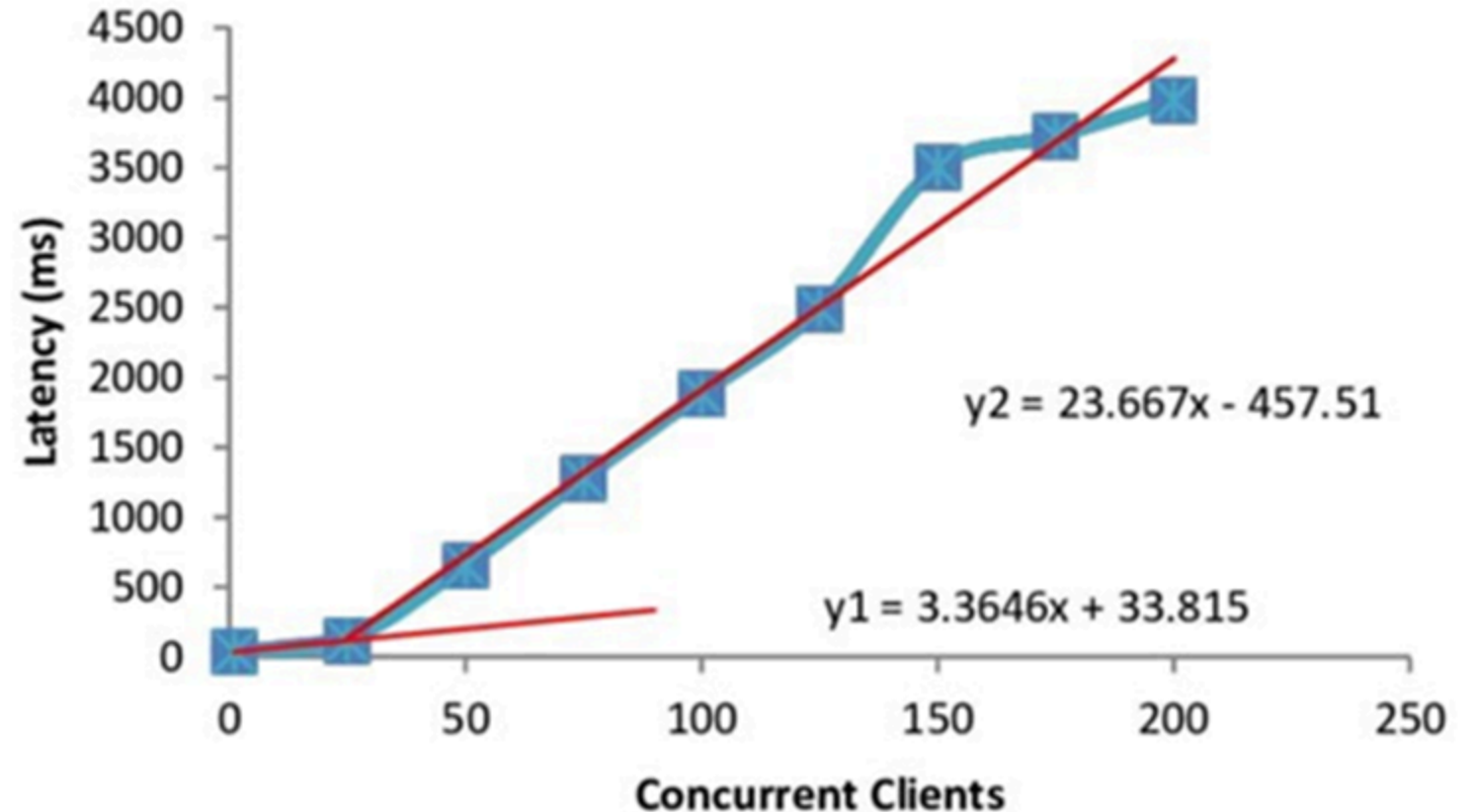
[Google Cloud Active Assist smart scheduling.](#)

Prospective solutions: cloud strategy

Three-way [trade-off](#):

1. Complexity: concurrent users → undesirable to target directly
2. Performance: latency, quality of service
3. Sustainability impact: required resources

... becomes trade-off between performance and sustainability!

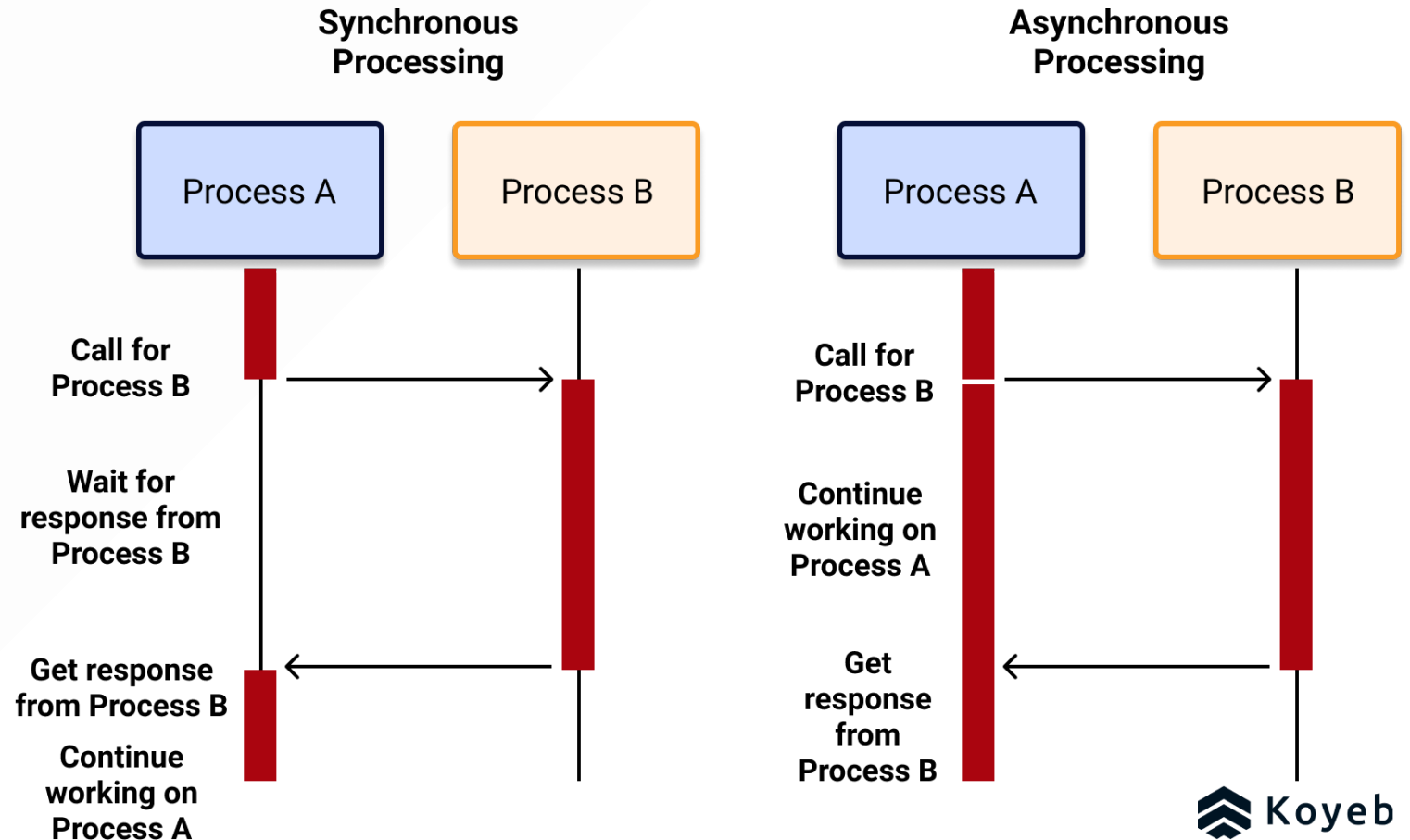


[Trade-off before bringing sustainability into the mix](#)

Prospective solutions: cloud strategy

Applying strategies that balance performance and sustainability:

- Adjust quality of result: [produce minimum needed computed output](#), remove unnecessary processing
- To maximize utilization, trade off response time by processing in batch, use [queues](#) for asynchronous processes
- Buffer and throttle to [flatten demand peaks](#)



[Synchronous and asynchronous processing.](#)

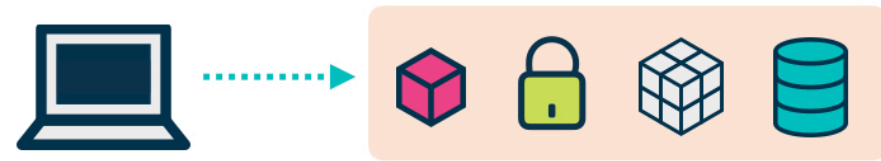
Prospective solutions: cloud strategy

Some strategies do not meaningfully affect performance:

- Switch from static stability (idle resources for redundancy) to [dynamic infrastructure scaling](#) (predictive scaling, schedule-based approach, serverless)
- Trade-off: computing activities closer to demand (edge computing) [reduce networking load](#), but closer to renewable energy reduce carbon intensity

TRADITIONAL vs SERVERLESS

TRADITIONAL



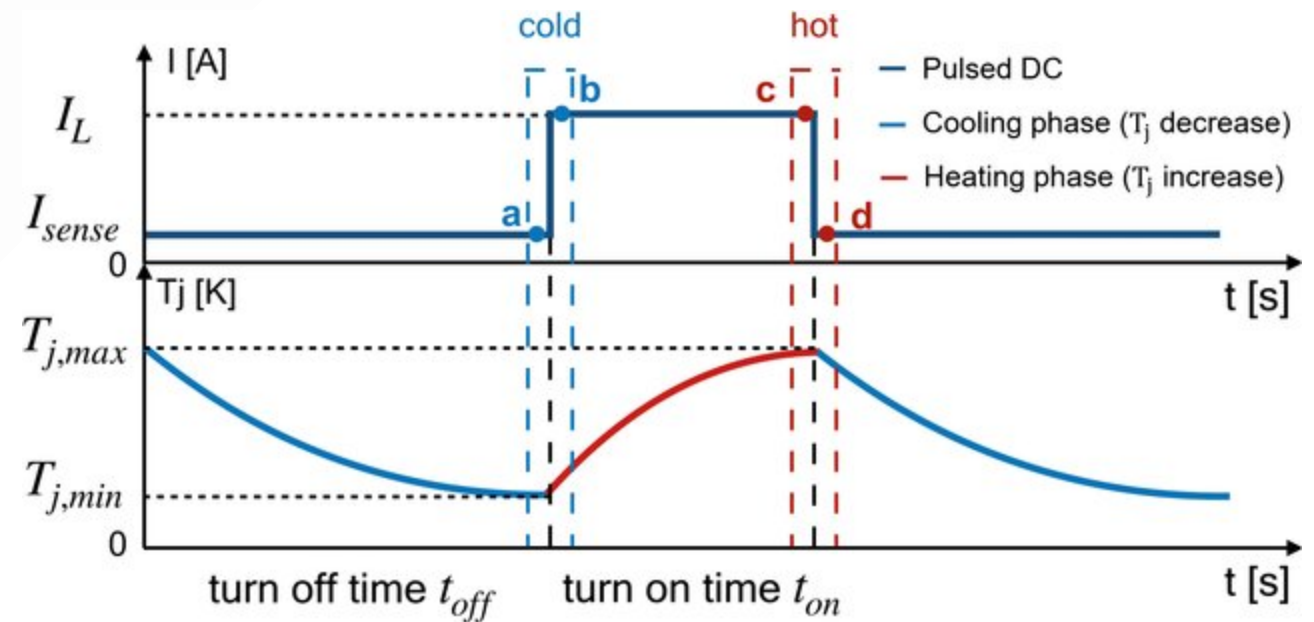
SERVERLESS (using client-side logic and third-party services)



'Serverless' just means someone else is managing it

Power cycling

- Smart scheduling: power cycling (on/off) of electronics
- Limited component lifetime: [thermal stress](#) due to expansion/compression
- Robust research on [lifetime estimation through power cycling](#), [lifecycle assessment of electronics](#), but none combining the two
- Complex trade-off: increase energy efficiency, but also frequency of component replacement...
- Master's thesis topic → contact me



[Component heating and cooling during power cycling.](#)

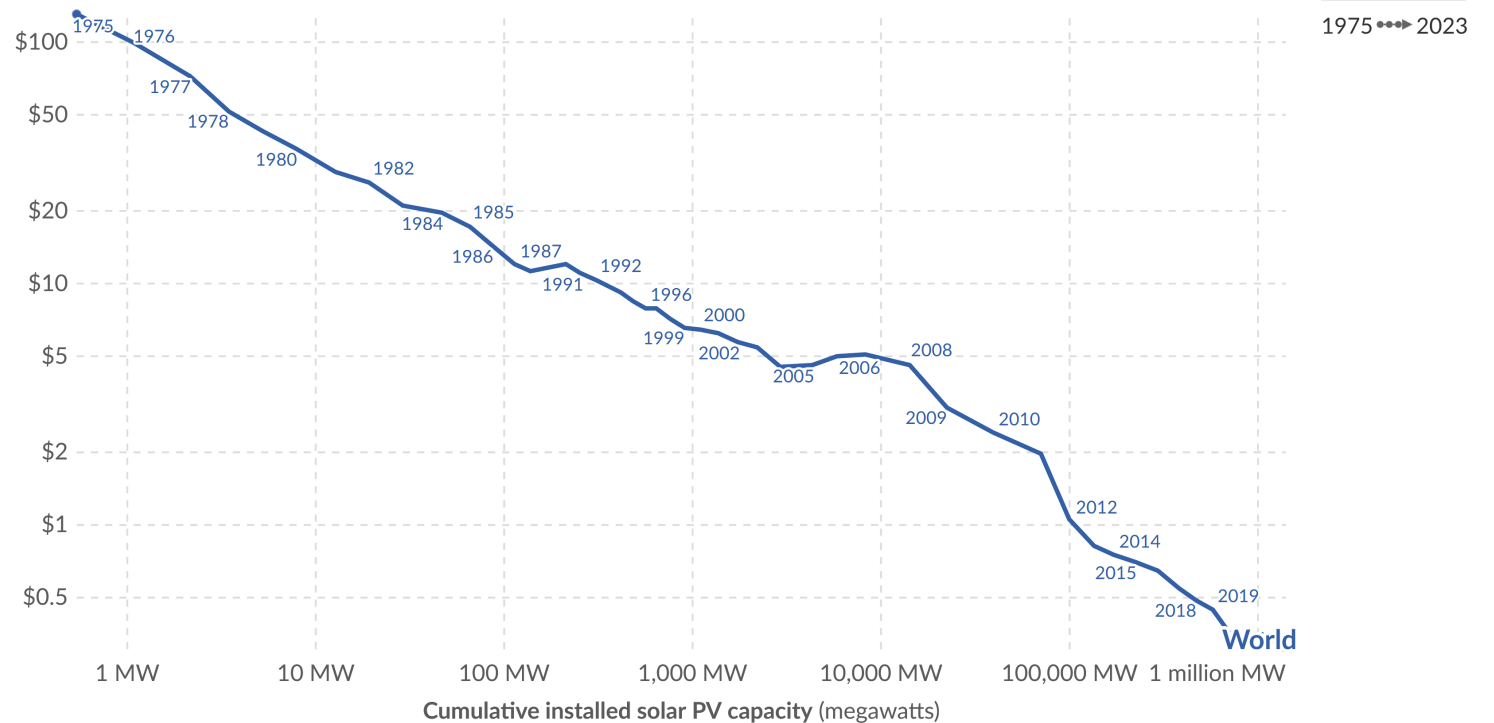
Prospective solutions: renewable energy

- Renewables (solar, wind etc) do not emit GHG's [during operation](#)
- [Intermittency](#): variable output depending on weather conditions
- Favorable [learning curves](#): costs for solar/battery systems falling
- Potential solution: demand response - align high utilization periods with windows of surplus renewable energy

Solar (photovoltaic) panel prices vs. cumulative capacity

Learning curve for solar panels. This data is expressed in US dollars per watt, adjusted for inflation. Cumulative installed solar capacity is measured in megawatts.

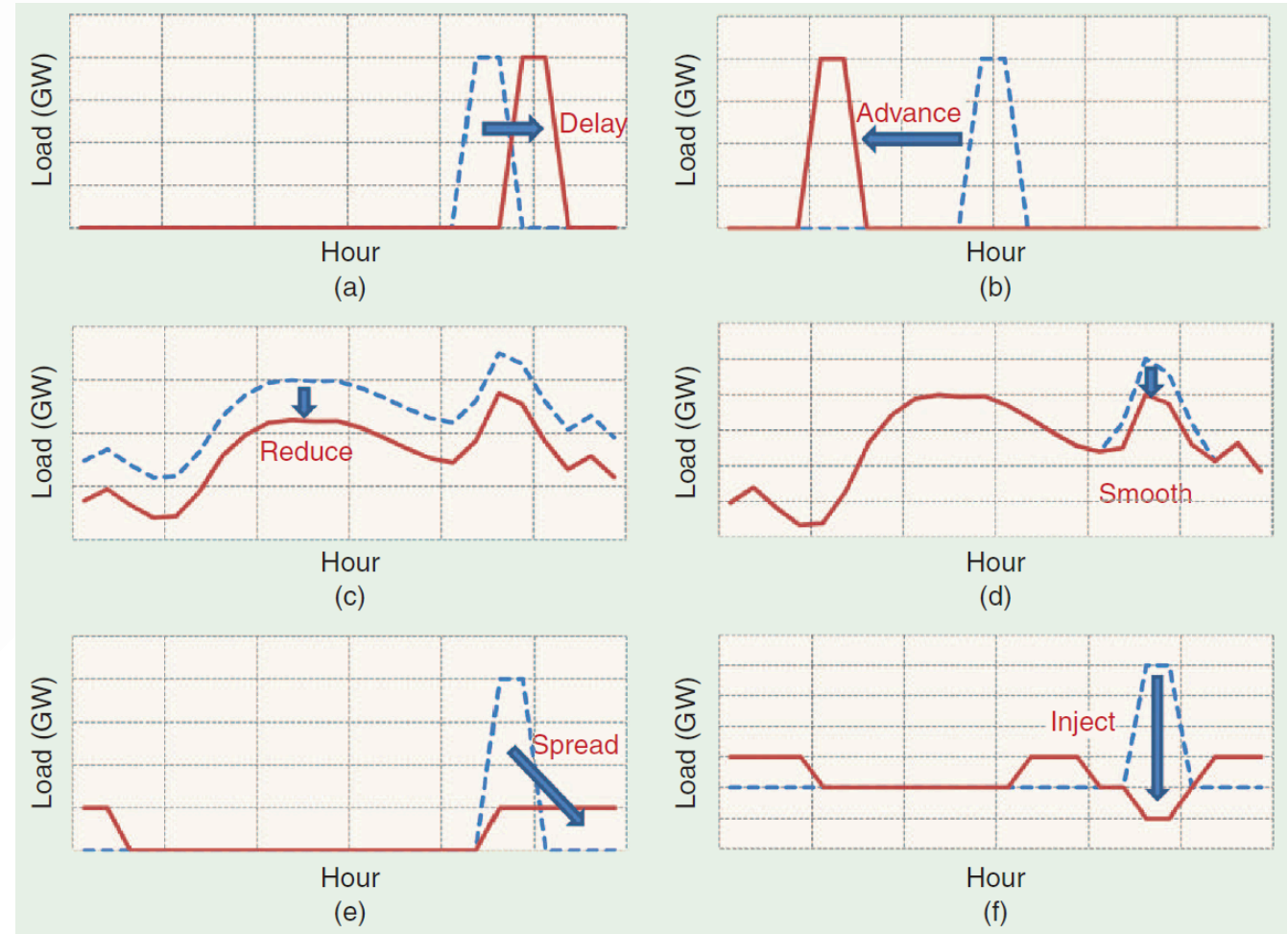
Solar PV module cost (constant 2023 US\$ per watt)



[Learning curve for solar PV - note logarithmic abscissa](#)

Prospective solutions: demand response

- Proposal: use parallelizable computing loads as variable demand to balance renewable energy grid
- [Example](#): Monte Carlo compute jobs (low quality of service, allow for pause, no strong privacy/security needs)
- Need to allow jobs to migrate from data center to data center depending on power supply → decentralized platform!



[Various demand response strategies](#)

Prospective solutions: server replacement

- Best practices in server maintenance → keep equipment for more than 3-5 years → fewer servers need to be produced
- Problem: latest generation hardware is more energy-efficient
- [Complex lifecycle trade-off](#): operate old, inefficient equipment or throw away and install new, efficient equipment

Prospective solutions: liquid cooling

- Always a trade-off between energy efficiency, water use
- [Novel cooling architectures](#): use liquids (higher heat capacity, [3000x dry air](#))
- Direct-to-chip (cold plate), immersive cooling: up to 50% more energy-efficient than air cooling
- Problem: direct use of water (closed-circuit?)
- Problem: toxic coolant (PFAS?)
- Higher costs, risks of flooding, difficult maintenance

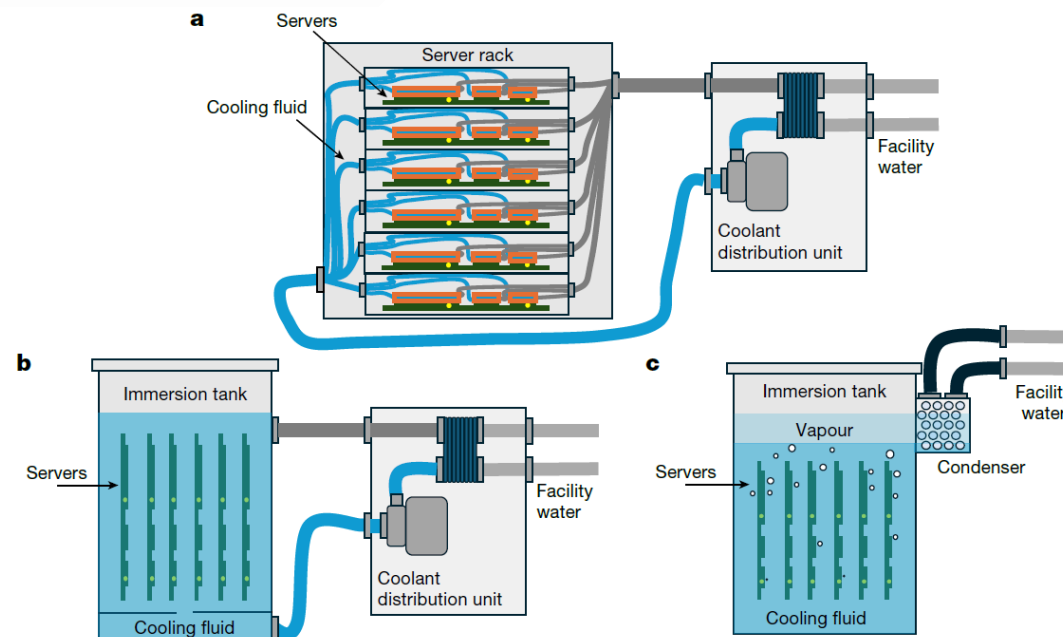


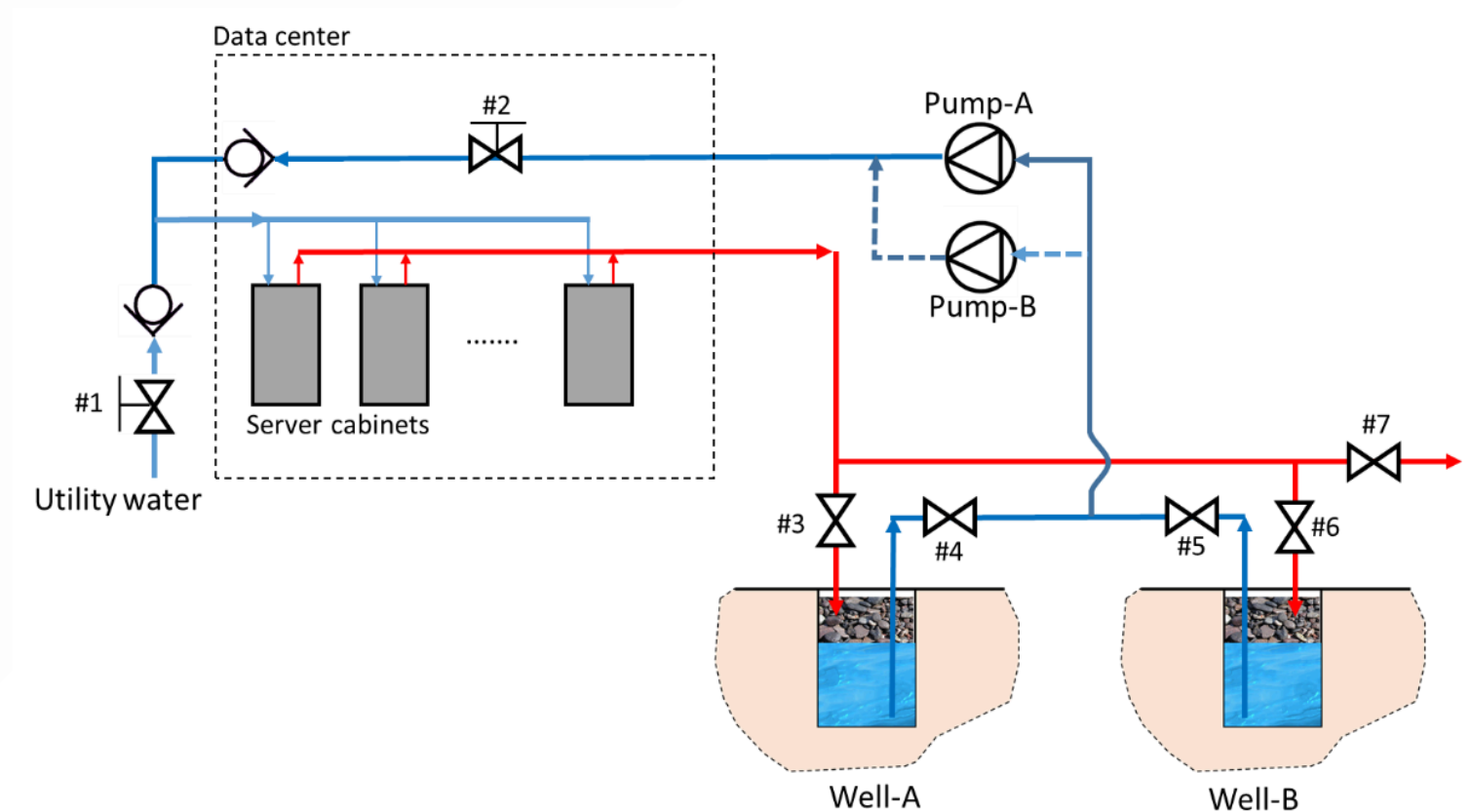
Fig. 1 | Liquid-cooling technologies provide alternatives to air cooling. Schematics showing the functional differences between the cooling technologies. **a**, Cold-plate cooling uses a metal plate with embedded channels for coolant, such as 25% polyethylene glycol and 75% water, to directly absorb and dissipate heat from electronic components with convection^{23,24}.

b, One-phase immersion systems immerse the entire server in a tank of dielectric fluid, such as mineral oil, to remove heat with convection²⁵. **c**, Two-phase immersion systems submerge servers in a dielectric fluid that boils at low temperatures, such as a refrigerant, which removes heat with the phase change²³.

[\(a\) cold plate](#), [\(b\) one-phase](#) and [\(c\) two-phase immersion](#)

Prospective solutions: geothermal cooling

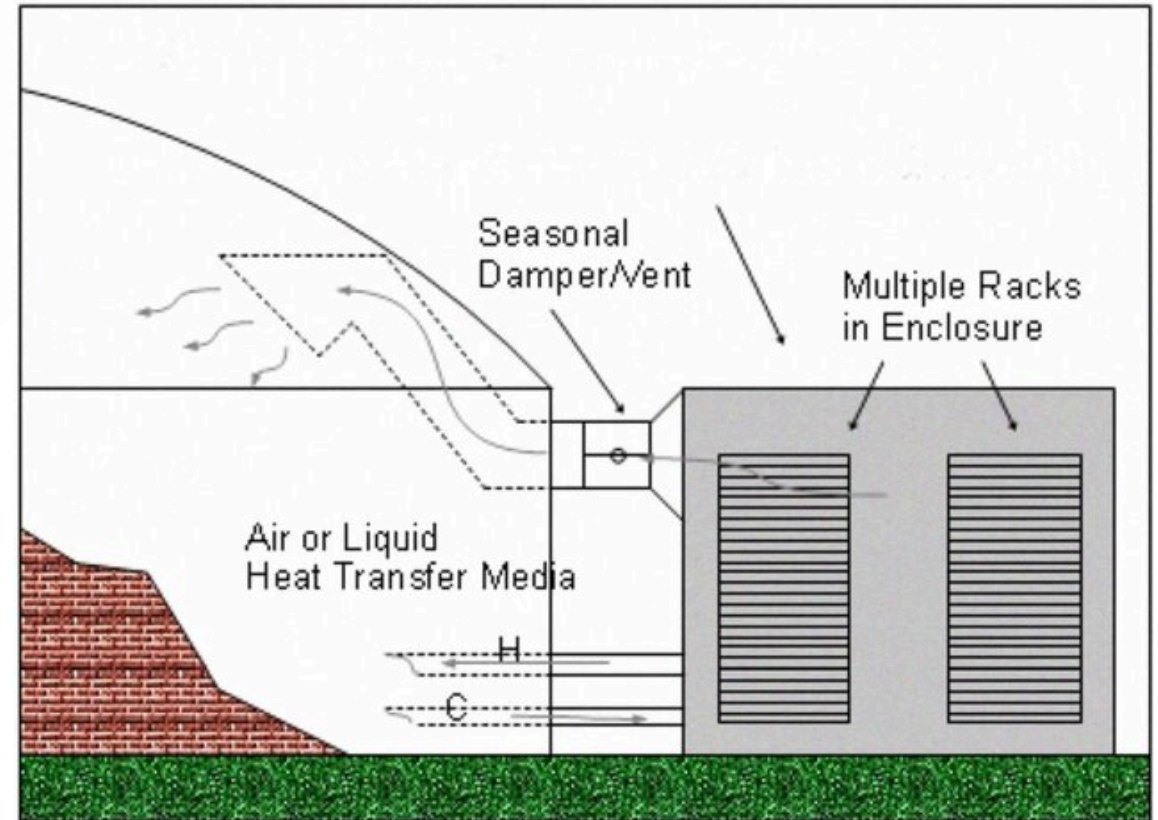
- 15-250 meters deep: [ground temperature](#) 10-15 degrees Celsius
- [Dig boreholes](#) and pump warm water from data center to cool down
- Example: Equinix, geothermally-cooled [Amsterdam](#) data center
- Lower operational costs, higher [installation costs](#)



[Geothermal cooling diagram with redundant cooling well](#)

Prospective solutions: waste heat reuse

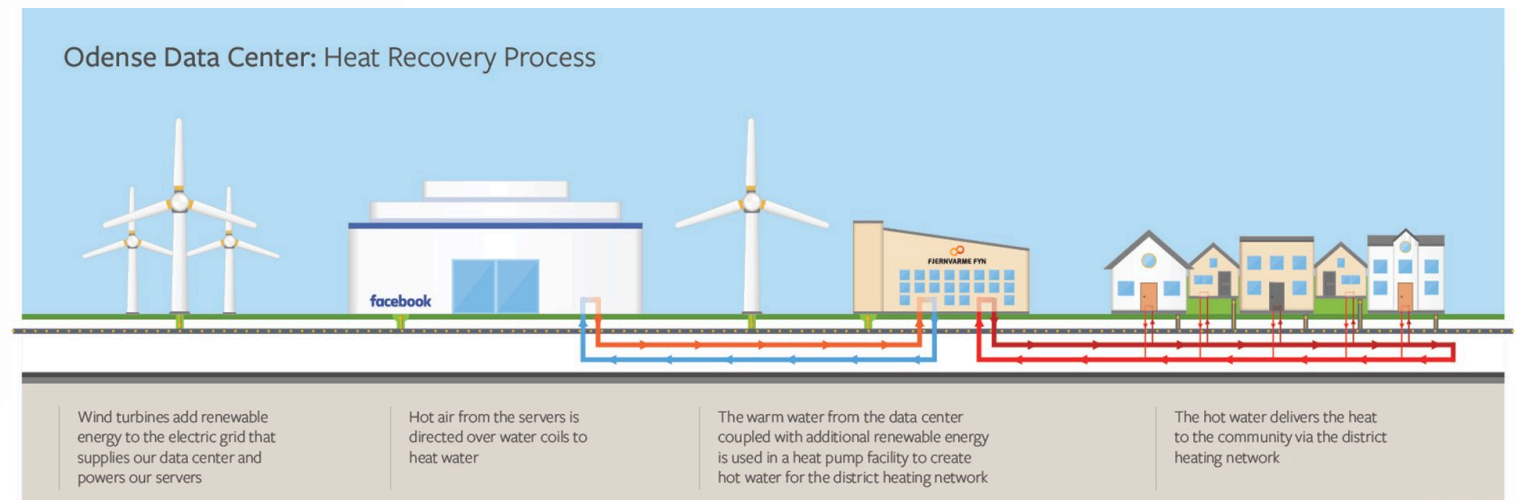
- Free heat → integrate with someone who has demand for heat: [industrial symbiosis](#)
- [Environmentally opportunistic computing](#) (EOC):
 - create heat where it is already needed
 - exploit cooling where it is already available
 - utilize energy when/where it is cheap
 - use mobility of virtualized services
- [Prototype](#): distributed container-based data center connected to greenhouse
- Computing jobs migrate depending on greenhouse temperature needs
- Dutch startup: [BlockHeating](#)



[Prototype distributed data center connected to greenhouse](#)

Prospective solutions: waste heat reuse

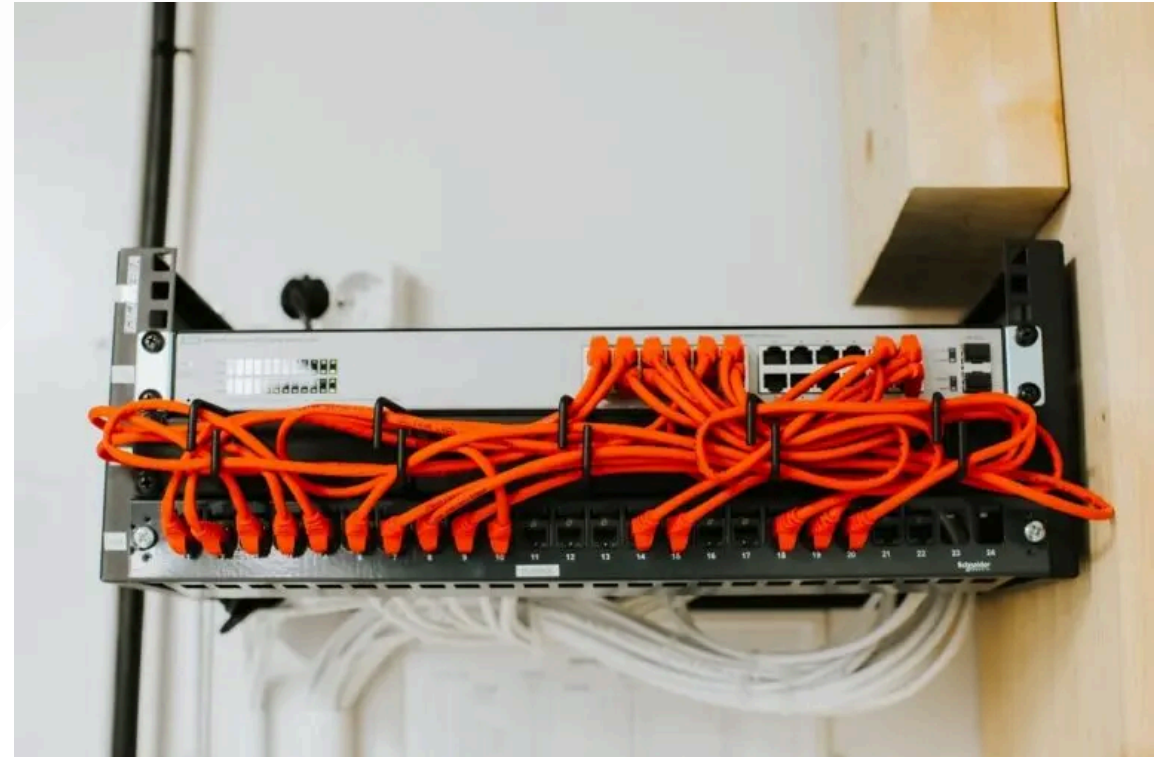
- Problem: heat demand profile (when do users need heat?)
- Problem: low quality heat produced (30-40 degrees), demand for high-quality heat (>50 degrees)
- [Odense, DK](#): Meta data center recovers waste heat, upgraded to high-quality heat, distributed by heat grid operator to buildings
- Similar developments pursued by [other tech companies](#)



[Heat recovery process in Meta's Odense data center](#)

Example: EcoDataCenter

- [EcoDataCenter](#): all-of-the-above approach to sustainability
- Based in Sweden: cold climate, free cooling
- 99.3% renewable energy (75% hydro, 25% wind), some backup use of diesel generators
- Waste heat delivered to district heating, industry needs
- Cooling systems can switch to dry cooling during droughts
- Water tanks used as thermal buffers
- Anything missing?



[EcoDataCenter's first building was constructed in wood](#)

Conclusions

- All solutions decrease energy/carbon/material intensity → efficiency gains, vulnerable to rebound effects
- Economic problem: business models of tech companies rely on (exponential) growth in abundance of digital services
- Finite planet → [energy constraints](#) to exponential growth?
- Technical solutions insufficient to contain exponential growth → economic/political solutions?

Recap

- Growth in projected data center compute/power demand, size, and fossil fuel use (!)
- Solutions for reducing energy/carbon/material intensity exist, but are limited and conflict with business models

Exam(ple) questions

1. Global annual carbon dioxide emissions reached 37.2 Gt in 2025. Assume 2% of these were caused by data centers, with the industry projected to grow by 5% and efficiency measures projected to reduce annual data center emissions by 2%. If global emissions grow by 1% each year, what percentage will be data center-related in 2050?
- a. 2.1%
 - b. 4.2%
 - c. 3.3%
 - d. 5.3%

Tip (don't expect this on the exam!):

$$\text{CO}_{2\text{final}} = \text{CO}_{2\text{initial}} \cdot \left(1 + \frac{\text{growth rate}}{100}\right)^{\text{years}}$$

Exam(ple) questions

1. Global annual carbon dioxide emissions reached 37.2 Gt in 2025. Assume 2% of these were caused by data centers, with the industry projected to grow by 5% and efficiency measures projected to reduce annual data center emissions by 2%. If global emissions grow by 1% each year, what percentage will be data center-related in 2050?

~~a. 2.1%~~

~~b. 4.2%~~

c. 3.3%

~~d. 5.3%~~

Tip (don't expect this on the exam!):

$$\text{CO}_{2\text{final}} = \text{CO}_{2\text{initial}} \cdot \left(1 + \frac{\text{growth rate}}{100} \right)^{\text{years}}$$

Feedback

- We need your feedback to continue improving the course!
- Feedback on Lecture 5 slides/notes:



Thank you for your attention!

Next lecture: Sustainable Electronics Design

Friday, October 2, 15:45-17:30, EEMCS Hall F

- Material basis of ICT systems, recycling
- Principles of sustainable economics design
- PCB design principles
- Battery-free electronics
- Digital product passports

Questions?